# Linux

Ubuntu - Through a DevOps Lens

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# Just Enough

# Linux

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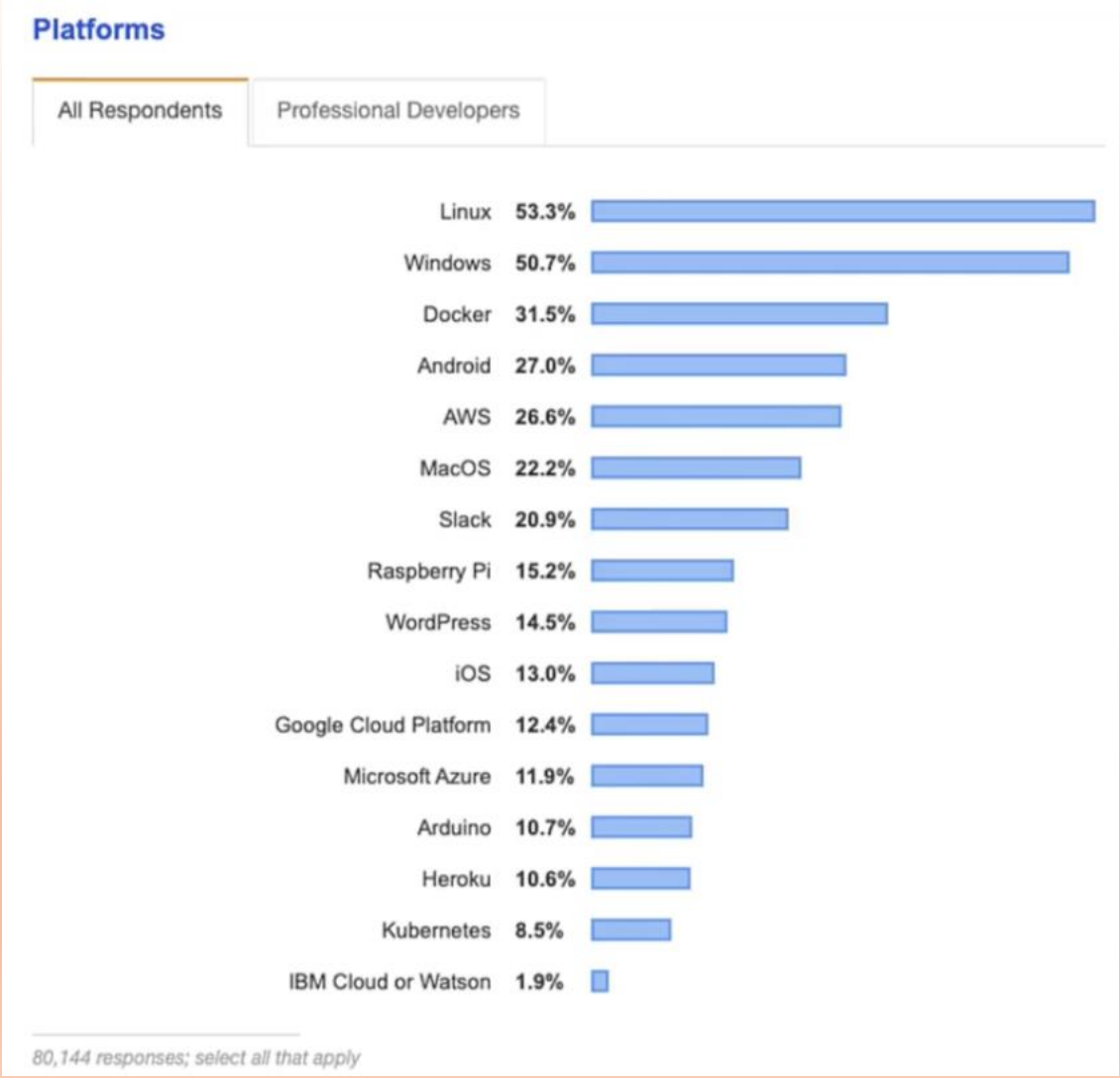
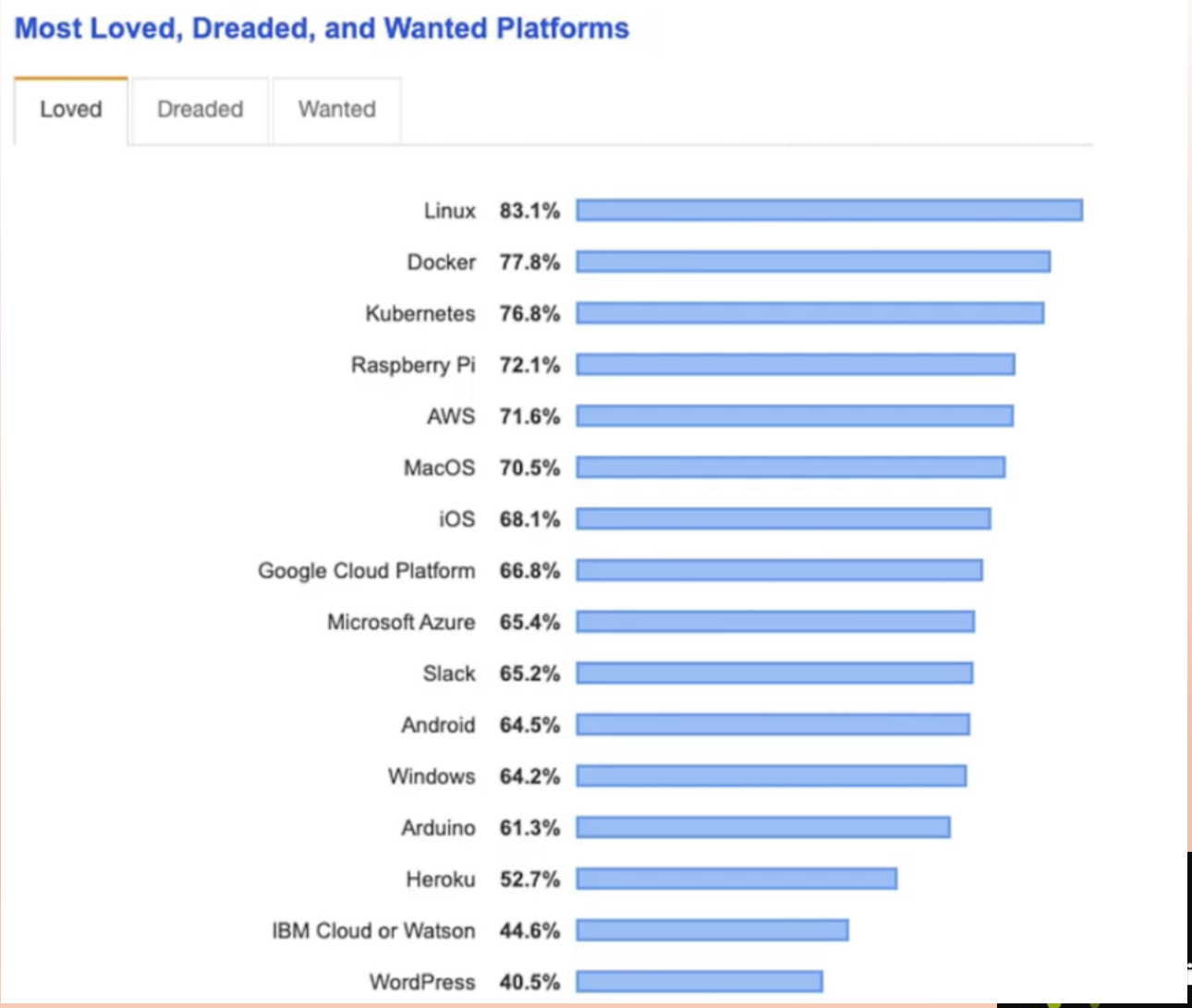
# Why Linux?

Developer Survey results

**2025**

# Why Linux?


Platforms
All Respondents
Professional Developers
Linux 53.3%
Windows 50.7%
Docker 31.5%
Android 27.0%
AWS 26.6%
MacOS 22.2%
Slack 20.9%
Raspberry Pi 15.2%
WordPress 14.5%
iOS 13.0%
Google Cloud Platform 12.4%
Microsoft Azure 11.9%
Arduino 10.7%
Heroku 10.6%
Kubernetes 8.5%
IBM Cloud or Watson 1.9%
80,144 responses; select all that apply


Most Loved, Dreaded, and Wanted Platforms
Loved
Dreaded
Wanted
Linux 83.1%
Docker 77.8%
Kubernetes 76.8%
Raspberry Pi 72.1%
AWS 71.6%
MacOS 70.5%
iOS 68.1%
Google Cloud Platform 66.8%
Microsoft Azure 65.4%
Slack 65.2%
Android 64.5%
Windows 64.2%
Arduino 61.3%
Heroku 52.7%
IBM Cloud or Watson 44.6%
WordPress 40.5%

# Why Linux?

2013 – Docker was born
2016 – Docker for Windows was born

## ✅ Can a Kubernetes master node run on Windows?

### No — Kubernetes master (control plane) cannot run on Windows.

The Kubernetes control plane **requires Linux** because its core components depend on Linux-specific primitives:

## ✅ Can Ansible run on Windows?

**Short answer:**

- **Yes, Ansible *can manage* Windows machines.**
- **No, Windows cannot natively run Ansible as a control node** (the machine that executes `ansible` and `ansible-playbook`).
- **But** you *can* run Ansible on Windows **indirectly** using WSL or Docker.

Let's break it down cleanly.

# Linux Principles

- Everything is a file (Including Hardware)
- Small Single purpose Programs
- Ability to chain programs together for complex operations
- Avoid Captive User Interface
- Configuration data stored in text file

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# Popular Linux distros

- Popular Desktop Linux OS
  - Ubuntu Linux
  - Linux Mint
  - Arch Linux
  - Fedora
  - Debian
  - OpenSuse
- Popular Server Linux OS
  - Red Hat Enterprise Linux
  - Ubuntu Server
  - Centos
  - SUSE Enterprise Linux

**Most used Linux distros currently in IT industry.**

## RPM based

- RHEL,
- Centos,
- Oracle Linux

## Debian based

- Ubuntu Server,
- Kali Linux

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Linux CLI

VI Editor

Package management

Service Management

# Shell Types

- Bourne Shell (sh Shell)
- C Shell (csh)
- Z Shell (zsh)
- Bourne again Shell (bash)

```
echo $SHELL
/bin/bash
```

# Basic Commands

```
echo Hi
Hi
```

Print to Screen

```
ls
File.txt my_dir1 file2.conf
```

Listing the files and directories

```
cd my_dir1
```

Change Directory

```
pwd
/home/my_dir1
```

Check Present working Directory

```
mkdir new_directory
```

Make/Create Directory

```
cd new_directory; mkdir www; pwd
/home/my_dir1/new_directory
```

Executing multiple commands in one line.

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# Commands - Directories

/test/Telangana/Hyderabad/Ameerpet/ACS

- mkdir /test
  mkdir /test/Telangana
  mkdir /test/Telangana/Hyderabad
  mkdir /test/Telangana/Hyderabad/Ameerpet
  mkdir /test/Telangana/Hyderabad/Ameerpet/ACS

- mkdir –p /test/Telangana/Hyderabad/Ameerpet/ACS

- rmdir –r /test/my_dir1

- cp –r my_dir1 /test/my_dir2

Make a directory hierarchy

Remove directory

Copy directory

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# Commands - Files

| | |
|---|---|
| touch new_file.txt | Create a new file (no contents) |
| cat > new_file.txt<br>Working with Linux<br>*CTRL+D* | Add contents to file |
| cat new_file.txt<br>Working with Linux | View contents of file |
| cp new_file.txt copy_file.txt | Copy file |
| mv new_file.txt sample_move.txt | Move (rename) File |
| rm new_file.txt | Remove File |

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# VI Editor

VI - Visual display editor
VIM - Visual display editor improved

```
atulm@Atuls-MacBook-Air Python % cat index.html
<!DOCTYPE html>
<html>
<body>
<h1> The VIM Poem <h1>

<p> The wise name said, just walk this way <p>
<p>To the dawn of the light<p>
<p>The wind will blow into your face<p>
<p>As the years pass you by<p>
<p>Hear this voice from deep Island<p>
<p>Its the call of your heart<p>
<p>Close your eyes and your will find<p>
<p>The passage out of the dark<p>
<p>V-I-M<p>
<p>Will you send me an angel<p>
<p>V-I-M<p>

<body>
<html>

atulm@Atuls-MacBook-Air Python %
```

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Command Mode

Insert mode (edit mode)

**Note: When you open the vi editor, it will be in the command mode by defa**

# User Accounts

- Every user of the system is assigned a unique user ID ( the UID)
- Users name and UID are stored in /etc/paswd
- Users passwords are stored in /etc/shadow in encrypted form
- Users are assigned home directory and a program that is run when they login (usually a shell)
- Users cannot read, write or execute each others files without permissions.

| TYPE | EXAMPLE | USER ID (ID) | GROUP ID (GID) | HOME DIR | SHELL |
|---|---|---|---|---|---|
| ROOT | root | 0 | 0 | /root | /bin/bash |
| REGULAR | Atul, ubuntu, | 1000 to 60000 | 1000 to 60000 | /home/username | /bin/bash |
| SERVICE | ftp, ssh, apache | 1 to 999 | 1 to 999 | /var/ftp etc | /sbin/nologin |

# Passwd file

```
ubuntu@ip-172-31-46-65:~$ cat /etc/passwd | grep ubuntu
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
```

| Field Number | Field Name | Value | Meaning |
|---|---|---|---|
| 1 | Username | ubuntu | The login name of the user |
| 2 | Password Placeholder | x | Indicates password is stored securely in /etc/shadow |
| 3 | UID (User ID) | 1000 | Unique ID for the user; first normal user gets UID 1000 |
| 4 | GID (Group ID) | 1000 | Primary group ID for this user |
| 5 | GECOS (User Info) | Ubuntu | Comment field; usually full name or description |
| 6 | Home Directory | /home/ubuntu | User's personal directory |
| 7 | Login Shell | /bin/bash | Default shell started when user logs in |

# Group file

```
ubuntu@ip-172-31-46-65:~$ head /etc/group
root:x:0:
daemon:x:1:
bin:x:2:
sys:x:3:
adm:x:4:syslog,ubuntu
tty:x:5:
disk:x:6:
lp:x:7:
mail:x:8:
```

**Group name, group password, GID, group members**

# ADD USER, SET PASSWORD & SWITCH TO USER

```
ubuntu@ip-172-31-42-204:~$ sudo adduser devops
info: Adding user `devops' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `devops' (1001) ...
info: Adding new user `devops' (1001) with group `devops (1001)' ...
info: Creating home directory `/home/devops' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for devops
Enter the new value, or press ENTER for the default
        Full Name []: DevOps User
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] Y
info: Adding new user `devops' to supplemental / extra groups `users' ...
info: Adding user `devops' to group `users' ...
ubuntu@ip-172-31-42-204:~$
ubuntu@ip-172-31-42-204:~$ su - devops
Password:
devops@ip-172-31-42-204:~$
devops@ip-172-31-42-204:~$
```

# ADD USER, GROUP & USER INTO GROUP

```
ubuntu@ip-172-31-46-65:~$ sudo groupadd DevOpsadmin
ubuntu@ip-172-31-46-65:~$
ubuntu@ip-172-31-46-65:~$ usermod -G DevOpsadmin DevOps
usermod: Permission denied.
usermod: cannot lock /etc/passwd; try again later.
ubuntu@ip-172-31-46-65:~$ sudo usermod -G DevOpsadmin DevOps
ubuntu@ip-172-31-46-65:~$ id DevOps
uid=1001(DevOps) gid=1001(DevOps) groups=1001(DevOps),1002(DevOpsadmin)
```

| Commands | Description |
| --- | --- |
| useradd | Creates user in RedHat |
| adduser | Creates user in ubuntu |
| id | shows user info |
| groupadd | Creates group |
| usermod –G grpname username | Adds user to a group |
| passwd | set/reset password |
| userdel –r | removes user with home-directory |
| groupdel | removes group |
| last | shows last login in system |
| who | who is logged in to a system |
| whoami | username |
| lsof –u user | list files opened by user |

# Sudo

- sudo gives power to a normal user to execute commands which is owned by root user.
- /etc/sudoers
- Elevated privileges – same as root/superuser
- User must be part of /etc/sudoers list

```
ubuntu@ip-172-31-46-65:~$ ls /root
ls: cannot open directory '/root': Permission denied
ubuntu@ip-172-31-46-65:~$ sudo ls /root
snap
```

# Download Files

- curl https://www.some-site.com/some-file.txt -o
- wget https://www.some-site.com/some-file.txt -o some-file.txt
  - -o to save the file otherwise file will be printed on the screen.

# Check OS version

- ls /etc/*release*

```
ubuntu@ip-172-31-46-65:~$ ls /etc/*release*
/etc/lsb-release  /etc/os-release
```

- cat /etc/*release*

```
ubuntu@ip-172-31-46-65:~$ cat /etc/*release*
DISTRIB_ID=Ubuntu
DISTRIB_RELEASE=24.04
DISTRIB_CODENAME=noble
DISTRIB_DESCRIPTION="Ubuntu 24.04.3 LTS"
PRETTY_NAME="Ubuntu 24.04.3 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.3 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
```

# The Role of Package Managers

- **Easy Software Management**

Package managers help you install, update, and remove software on Linux easily, making things faster and simpler.

- **Handles Software Needs Automatically**

They take care of all the extra software your programs need, so nothing breaks or goes missing.

- **Keeps Your System Safe and Easy to Manage**

Using one tool to manage everything makes your system safer and saves you time compared to doing it all by hand.

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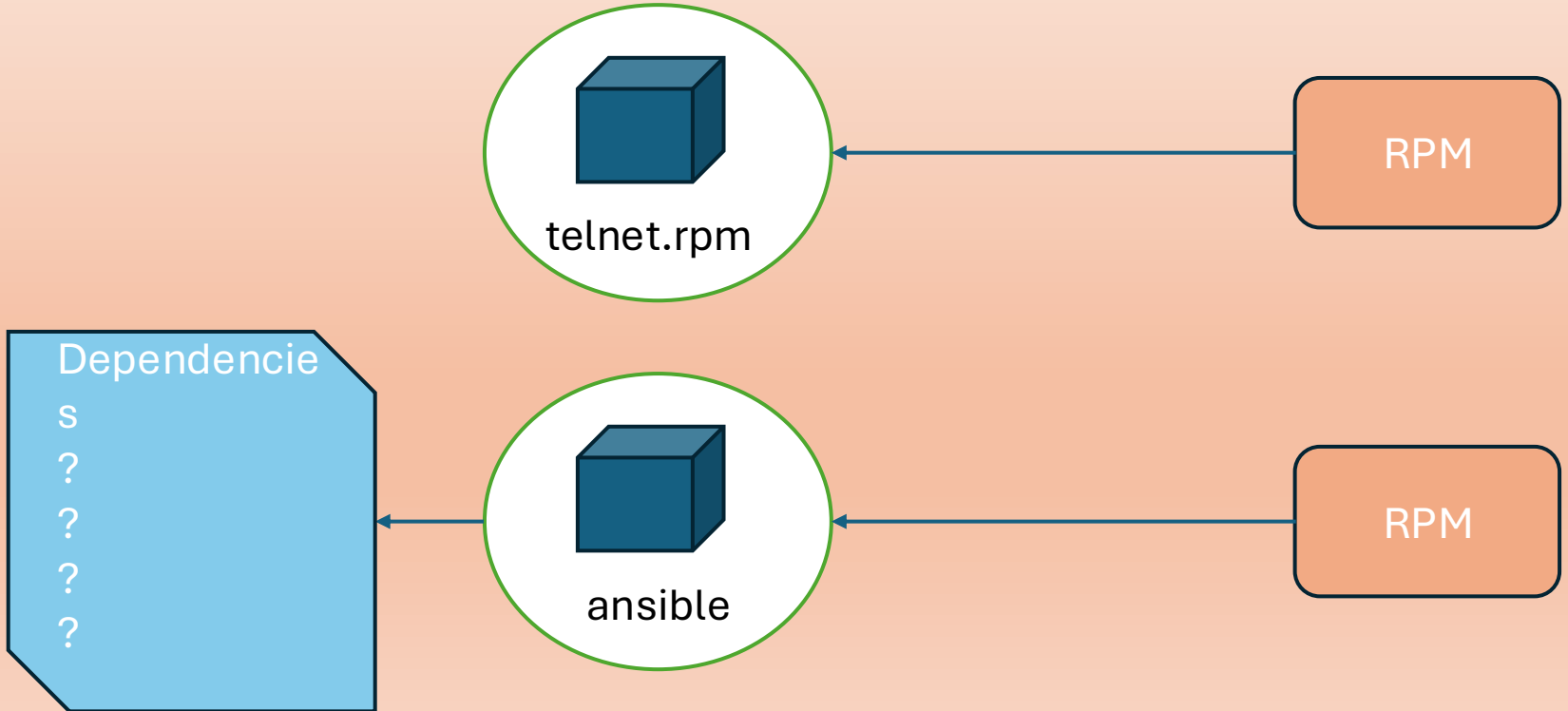
# RPM (RedHat Package Manager)

- *rpm -i some-package.rpm*
- *rpm –e some-package.rpm*
- *rpm –q some-package.rpm*

Install a package

Uninstall package

query package



# YUM (Yellowdog Updater, Modified)

- YUM is a package manager used in RHEL-based Linux distributions (CentOS, Red Hat, Oracle Linux, Amazon Linux).
- High-level package manager and uses RPM underneath
- Key Functions
  - Install packages with automatic dependency resolution
  - Update system packages to latest versions
  - Remove unwanted packages cleanly
  - Search for available software
  - Manage repositories (enable/disable/add repos)
  - Handle dependencies without manual intervention.

# Common YUM commands

| Task | Command |
|---|---|
| Install a Package | yum install httpd |
| Update System | yum update |
| Remove Package | yum remove httpd |
| Search Package | yum search nginx |
| List installed Packages | yum list installed |
| Clean Cache | yum clean all |

# APT (Advanced Package Tool)

- APT is the primary package management system used in Debian and its derivatives (Ubuntu, Mint, Kali, etc.).
- APT (Command-line frontend)
  - High-level tool used for:
  - Installing packages
  - Updating repositories
  - Upgrading the system
  - Removing packages
    - apt update
    - apt install nginx
    - apt upgrade

# Continued...

- dpkg (Low-level package manager)
  - Installs or removes .deb files directly
  - Does NOT handle dependencies automatically
  - *Example: dpkg -i package.deb*
- APT-GET (Older CLI tool)
  - Predecessor to apt
  - Still widely used in scripts
  - *Example: apt-get install apache2*

Yes/No → confirmation

# Ubuntu Commands

| Command | Purpose |
|---|---|
| `apt update` | Refresh package index |
| `apt search <PACKAGE>` | Search for packages |
| `apt install <PACKAGE> -y` | Install a package |
| `apt reinstall <PACKAGE>` | Reinstall a package |
| `apt install --only-upgrade <PACKAGE>` | Upgrade a single package |
| `apt remove <PACKAGE>` | Remove package |
| `apt purge <PACKAGE>` | Remove package + config |
| `apt autoremove` | Remove unused dependencies |
| `apt clean` / `apt autoclean` | Clean cache |
| `apt list --installed` | List installed packages |
| `apt show <PACKAGE>` | Show package details |
| `apt depends <PACKAGE>` | Show dependencies |
| `apt rdepends <PACKAGE>` | Reverse dependencies |
| `apt --fix-broken install` | Fix broken installs |

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# Service

- In Linux, services help manage software to run continuously in the background, ensuring they start automatically during server reboots and follow the proper startup sequence.
- When software like web servers, database servers, or DevOps tools such as Docker are installed, they are typically set up as services on the system by default.
- The modern way to start a service is by using the system kernel command, specifically the systemctl command, followed by the service name.
- On current systems, the SERVICE command functions as a wrapper around systemctl.

# Services

| | |
|---|---|
| sudo service httpd start | Start httpd service (old) |
| sudo systemctl start httpd | Start httpd service (new) |
| sudo systemctl stop httpd | stop httpd service |
| sudo systemctl status httpd | check httpd service status |
| sudo systemctl enable httpd | Configure httpd to start at startup |
| sudo systemctl disable httpd | Configure httpd to not start at startup |

***Note: httpd in this example is an Apache web server.***

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# how do you configure a program or software as a service?

etc/systemd/system

- Creating a systemd unit file
- Defining how your application starts, stops, restarts
- Defining environment variables, working directory, user, logs
- Placing the service file in the correct directory
- Reloading systemd so it recognizes your service

*#This is the core of "configuring a service."*

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# Continued…

etc/systemd/system

- creating a systemd unit file
  - sudo vi /etc/systemd/system/hello-web.service

- #Put this inside (adjust paths if your username or directory differs):

```
[Unit]
Description=Simple Python Hello World Web Service
After=network.target
[Service]
User=ec2-user
WorkingDirectory=/home/ec2-user/hello-web
ExecStart=/home/ec2-user/hello-web/venv/bin/python /home/ec2-user/hello-web/app.py
Restart=always
Environment="PATH=/home/ec2-user/hello-web/venv/bin"
Environment="PYTHONUNBUFFERED=1"
[Install]
WantedBy=multi-user.target
```

# Continued....

- #4. Reload systemd and start the service

```
sudo systemctl daemon-reload
sudo systemctl start hello-web
sudo systemctl status hello-web
```

- If status shows active (running), you're good.

- Enable it at boot:

```
sudo systemctl enable hello-web
```